

Taitil Chheda

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Education

Dwarkadas J. Sanghvi College of Engineering
B.Tech Artificial Intelligence & Machine Learning
Grade Awarded - 8.66/10

Batch - 2027

Internships

Eccomix - AI ML Developer

April 2024 - August 2024

- Used **PyMuPDF** to scan and extract the pages of pdf to image for further image processing.
- Compared multiple different **OPEN SOURCE LLMs** and found one that balances speed/CPU utilization.
- Improved the previous prompts and made the **RAG model** robust.

Robokart - Robotics Trainer

June 2024 - July 2024

- Provided insights on **emerging technologies**, including **AI, IoT, and robotics**, enhancing students' technological proficiency.
- Led robotics-based projects across institutes in Gujarat under the **Gujarat Government, GUJCOST, DST and KCG**.
- **Recognized in media** for impactful work in science and technology education with the Gujarat government.

Projects

Emotion Recognition - An AI-based Emotion Detection Tool

- Developed a custom emotion recognition system using (**CNN**) and **Computer Vision** techniques.
- Trained model to accurately detect emotions like happiness, sadness, and surprise from facial expressions.
- Integrated custom **CNN** model for efficient face **feature extraction** and emotion classification.
- Achieved high accuracy through meticulous **fine-tuning** and experimentation.

Technologies Used: **Python, TensorFlow, OpenCV, Convolutional Neural Networks (CNN), Computer Vision.**

QueryGenius- A Machine Learning RAG/ chat bot that Answers Questions

- Led the development of an advanced **Language Model (LLM)** based Question Answering (QA) system using Hugging Face's **llama-index** library within the field of **Natural Language Processing (NLP)**.
- Engineered a state-of-the-art embedding model using **FastEmbed** to represent documents in a high-dimensional **semantic** space, enhancing the system's ability to understand and retrieve relevant information.
- Configured and optimized the **Gemma** variant of the GPT architecture from **Google's "google/gemma-7b-it" model**, specializing in language comprehension for improved accuracy.
- Developed a **user-friendly** chat interface using **Gradio**, facilitating seamless interaction with the **LLM-based QA system**.

Technologies Used: **Hugging Face, llama-index, FastEmbed, Transformers, Gradio, PyTorch, Gemma language model.**

Stock Market Prediction- A Machine Learning Model that predicts stock prices

- Developed a ML model to predict **NIFTY ('NSEI') Index & stock prices**, enhancing insight into market trends.
- Utilized Python and the **yfinance** library to gather and preprocess historical stock data.
- Engineered features and trained a **linear regression model** to forecast next day's closing price.
- Evaluated model accuracy using **Mean Squared Error (MSE)** and **R-squared metrics**.
- Visualized historical and predicted prices, leveraging **matplotlib** for impactful data storytelling.

Technologies Used: **Machine Learning, Python, finance, pandas, scikit-learn, matplotlib.**

SAR Colorizer- A Deep learning model to colourize SAR images using GANs.

- Led the development of an advanced Generative Adversarial Network (**GAN**) for colorizing **Synthetic Aperture Radar Images**, transforming grayscale SAR data into **realistic color images**.
- Developed a custom **U-Net generator** to predict color images from grayscale SAR inputs, using a multi-scale approach and **discriminator** for enhanced output quality and local discrimination.
- Optimized the model's performance using **TensorFlow** and **Keras**, implementing a **hybrid loss function** combining adversarial loss and L1 loss to balance perceptual and pixel-wise accuracy.

Technologies Used: **TensorFlow, Keras, Generative Adversarial Networks, U-Net, OpenCV, Deep Learning**

Skills

- **Tech Stack:** Python, Machine Learning, Deep Learning, LLM (large language model), NLP (Natural Language Processing), Data Modeling and evaluation, Tensorflow, Numpy, OpenCV, SkLearn, Matplotlib, PyTorch, Pandas, Keras, Transformers, Image Processing, HTML, CSS, Java, C, DSA, GANs (Generative Adversarial Networks)
- **Non Tech Skills:** Creative Problem-Solving, Critical & Analytical Thinking, Leadership, Excellent Communication skills, Decision-making, Adaptability, Playing Keyboard, Photography, Gaming, Chess, Table tennis, Painting.